MILITARY SPECIFICATION

HOSE, RUBBER, PETROLEUM BASED FUELS AND WATER SERVICES, DISCHARGE ONLY, SMOOTH BORE, LIGHTWEIGHT BUOYANT TYPE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight, smooth bore, petroleum based fuels and water service discharge rubber hoses.

1.2 Classification.

1.2.1 Hose designation. Hose covered by this specification shall be designated in the following form (see 6.2.1).

M22240-XX-X

Military specification code number
Hose size code number (see 1.2.1.1)
Hose type identification code letter (see 1.2.1.2)

1.2.1.1 Hose size. The hose size (inside diameter) is identified by two digits:

Size code number	Inside diameter (inches)
25	2-1/2
40	4
60	6
70	7

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.2 Hose type. Hose type is identified by a single capital letter.

Type code letter	Type	Sizes
A	Alongside fueling	40, 60, and 70
B	Astern fueling	60 only
C	Alongside fueling or water transfer	25

1.3 Water services include both boiler feed water and potable water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-S-735 - Standard Test Fluids, Hydrocarbon.

MILITARY

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers.

MIL-C-24356 - Couplings, Segmented, Reattachable, 2-1/2-, 4-, 6- and 7-inch, for Refueling-at-Sea Hose.

STANDARDS

FEDERAL

FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.

2.1.2 Government documents. The following Government document forms a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations
Federal Food, Drug and Cosmetic Act
21 CFR 177.2600

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 297 - Rubber Products - Chemical Analysis, Methods for. (DoD adopted)
D 380 - Rubber Hose, Methods of Testing.
D 381 - Existent Gum in Fuels by Jet Evaporation, Test Method for. (DoD adopted)
D 412 - Rubber Properties in Tension, Test Method for. (DoD adopted)
D 413 - Rubber Property - Adhesion to Flexible Substrate, Test Method for. (DoD adopted)
D 573 - Rubber Deterioration in an Air Oven, Test Method for. (DoD adopted)
D 2240 - Rubber Property - Durometer Hardness, Standard Test Method for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction - types A, B, and C. The hose shall have a rubber tube, reinforcement consisting of either braided synthetic fiber yarn (type C only) or spiral plies of synthetic fiber cord (four plies minimum for type A) fully bonded together with rubber, friction impregnated through the cord so as to create a rubber-to-rubber bond, and a rubber cover. All layers shall be bonded to their adjacent layers so as to provide a continuous rubber bond from the inner tube through the reinforcement to the outer cover. Vulcanization shall be uniform over the length of the hose. Ends of the hose shall be cut square and shall not be capped. Synthetic fiber cord shall be polyamide, or polyester. Types A, B, and C shall be compatible with couplings in accordance with MIL-C-24356.

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3.1.1 Type B only. In addition to the construction requirements specified in 3.1, type B hose shall be reinforced longitudinally in such a manner to meet the requirements of 3.8.1 when tested in accordance with 4.8.9.1.

3.2 Appearance. The tube, reinforcement and cover shall have no more total visual defects than are allowed in 4.5.1 when the inside surface, outside surface and ends of the hose are examined in accordance with 4.7.

3.3 Length. Unless otherwise specified (see 6.2.1), types A and C hose shall be 35 feet long, and type B hose shall be 50 feet long when measured in accordance with 4.8.1. The acceptable variation in length shall be plus or minus 1 percent (see 4.6.1). Residual test lengths at least 30 feet long are acceptable if inspected by the contracting officer (see 6.2.1).

3.4 Inside diameter, wall thickness, unit weight, and tube and cover thicknesses. These features of the hose when measured in accordance with 4.8.2 through 4.8.5 respectively, shall conform to the requirements listed in table I.

TABLE I. Dimensions, unit weight and test pressures.

	2-1/2 inch size	4-inch size
Inside diameter, inches	2.500 ± 0.063	4.000 + 0.062 - 0.000
Wall thickness, inch	0.312 ± 0.047	0.328 ± 0.047
Tube thickness, inch	0.063 min	0.094 min
Cover thickness, inch	0.063 min	0.078 min
Unit weight, lbs/ft	2.0 max	3.0 max
Proof pressure, lb/in^2	300	300
Burst pressure, lb/in^2	600	600 min
	6-inch size	7-inch size
Inside diameter, inches	6.000 + 0.125 - 0.000	7.000 + 0.125 - 0.000
Wall thickness, inch	0.422 ± 0.047	0.438 + 0.047 - 0.024
Tube thickness, inch	0.094 min	0.110 min
Cover thickness, inch	0.078 min	0.094 min
Unit weight, lbs/ft	6.0 max	7.0 max
Proof pressure, lb/in^2	300	300
Burst pressure, lb/in^2	600 min	600 min

3.4.1 Type B, 6-inch size only may vary from table I, as follows:

Weight - 6.3 pounds per foot (lbs/ft) maximum.
Wall thickness - 0.475 ± 0.047 not more than two locations 180 degrees apart.

3.5 Proof pressure. The hose shall withstand the proof pressure specified in table I without leakage or other indication of weakness when tested in accordance with 4.8.6.

3.6 Dimensional stability. When tested in accordance with 4.8.7, the hose shall not increase in length more than 10 percent or increase in outside diameter more than 18 percent, nor shall the twist be more than 1 degree per foot while under the proof pressure, and shall not retain more than 2 percent

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change in length or 3 percent change in outside diameter after releasing the pressure. All calculations shall be based on the original dimensions of the hose.

3.7 Burst pressure. The hose shall withstand the burst pressure specified in table I without bursting, leaking, or loosening or loss of the coupling when tested in accordance with 4.8.8.

3.8 Tensile pull. When tested in accordance with 4.8.9, the hose shall withstand the tensile pull listed in table II without any apparent damage to the hose or weakening of the coupling grips.

TABLE II. Tensile pull for hoses.

Size (inches)	Tensile pull (pounds)
2-1/2 (Type C)	5,000
4 (Type A)	5,000
6 (Type A)	12,000
7 (Type A)	12,000
6 (Type B)	16,000

3.8.1 Elongation and neckdown (type B). When tested in accordance with 4.8.9.1, type B hose shall have a maximum elongation of 7.5 percent and a maximum neckdown of 4.5 percent.

3.9 Rubber components. The properties of the rubber in the tube, cover, and friction of the hose shall conform to the requirements listed in table III. Suggested formulations for these components are given in 6.3.

3.9.1 Tube and friction. The rubber used in the tube and friction shall be a copolymer of butadiene and acrylonitrile. The tube shall be smooth, free of seams with any depth profile, and shall be of uniform thickness. Mandrel marks or scores present in the tube must exhibit a good molded appearance and indicate no evidence of tearing. A few indentations arising from air entrapment between mandrel and tube may be accepted provided such minor indentation has rounded edges and root, is not deeper than 10 percent of the minimum tube thickness, is firm and has full tube thickness.

3.9.1.1 Inner tube, type C hose. The inner tube material shall be compounded to conform to Federal Food, Drug and Cosmetic Act, 21 Code of Federal Regulations 177.2600. The contractor shall supply a certificate of compliance.

3.9.1.2 Water resistance. When tested in accordance with 4.8.11.10, the extractable non-volatile matter for the type C tube only shall not exceed 21 milligrams per square inch.

3.9.2 Cover. The rubber used in the cover shall be a blend or a copolymer of butadiene, acrylonitrile and a polyvinylchloride.

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TABLE III. Requirements for rubber components in the hose.

	Tube	Cover	Friction	Test paragraph
Initial properties:				
Tensile strength, lb/in^2, min	2300	1600	1800	4.8.11.1
Ultimate elongation, percent, min	300	350	400	4.8.11.1
Adhesion, lb/in, min	40	40	40	4.8.11.5
Properties after immersion in test fluid:				
Tensile strength, lb/in^2, min	1400	800	900	4.8.11.2
Ultimate elongation, percent, min	200	200	260	4.8.11.2
Volume increase, percent, max (no shrinkage permitted)	30	30	30	4.8.11.4
Adhesion, lb/in, min	25	25	25	4.8.11.6
Properties after oven aging:				
Tensile strength, lb/in^2, min	2000	1450		4.8.11.3
Ultimate elongation, percent, min	250	250		4.8.11.3
Nonvolatile extract from tube, percent, max	3.0			4.8.11.7
Phosphate plasticizer in tube, qualitative determination	Absent			4.8.11.8
Resistance of cover to ozone	No cracking			4.8.11.9
Specific gravity	Record	Record	Record	4.8.11.11
Durometer hardness	Record	Record	Record	4.8.11.12

3.10 Configuration control. Upon satisfactory completion of the first article inspection of 4.3 and acceptance of the first article by the Government, the hose materials, design, construction, and manufacturing technique shall not be altered chemically or physically without the prior approval of the Government. The first article shall be delivered to the Government upon completion of the first article examination and testing.

3.11 Branding and marking.

3.11.1 Hose. When inspected in accordance with 4.4.1.1, each length of hose shall be branded in letters at least 1/4 inch high inlaid in the cover and vulcanized thereto about 5 feet from each end. Branding shall include the following:

(a) Contractor's name or trademark.
(b) Month and year of manufacture.
(c) Specification number and revision letter.

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(d) National stock number (to be furnished by the command or agency concerned).
(e) Nominal hose size.
(f) Hose, types A, B, and C shall be branded as follows:

(1) Type A shall be marked "Alongside Fueling Only".
(2) Type B shall be marked "Astern Fueling Only".
(3) Type C shall be marked "Fuel or Water Service".

3.11.2 Striping. When inspected in accordance with 4.4.1.1, each length of hose shall be marked with a longitudinal white or red stripe along its entire length to show the natural relaxed position of the hose. This stripe shall be a white or red rubber compound securely vulcanized to the outer cover of the hose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all the requirements of section 3. The inspections set forth in this specification shall become part of the contractor's overall inspection or quality control program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all product or supplies submitted to the Government for acceptance comply with all requirements of this specification. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article shall be a first production item of the hose supplied under this specification. The first article inspection shall consist of the examination and tests specified in 4.7 and 4.8.1 through 4.8.11.

4.3.1 Rejection of first articles. Upon completion of the examination of the first article per 4.7, the presence of one or more defects shall be cause for rejection. Failure of any one test specified in 4.3 shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in 4.4.1, 4.7, 4.8.1 through 4.8.9 and 4.8.11.

4.4.1 Individual hose tests. Each hose or hose assembly shall be subjected to the following tests:

4.4.1.1 Visual examination. The exterior surfaces, interior surfaces, ends and fittings (if required) shall be examined for critical or major visual defects per 4.7. Any hose containing major or critical defects shall be rejected. Hose branding and marking shall be examined for conformance to 3.11. Hoses incorrectly marked or branded shall be rejected.

4.4.1.2 Proof pressure. Each hose or hose assembly shall be subjected to the applicable proof pressure as specified in table I for a minimum of 2 minutes. If the contract specifies hose with fittings (hose assembly), these fittings will be installed prior to the proof pressure test. If the contract does not include fittings on the hose, any fitting which will contain the proof pressure may be used. When temporary fittings have been installed and removed, there may be marks on the cover due to the fittings. These marks will not be cause for hose rejection. Dimension stability tests are not required, as they are covered in the lot sampling tests. Any hose exhibiting defects due to the proof pressure shall be rejected.

4.5 Sampling for quality conformance inspection.

4.5.1 Sampling for examination, measurements, and tests performed on entire hose. A lot shall consist of all hoses of the same type and size, manufactured under the same conditions and offered for delivery at one time. A random sample of hoses shall be selected from each lot in accordance with general inspection level II in accordance with MIL-STD-105, with Acceptance Quality Level (AQL) of 1.5 percent defective, and shall be subjected to the examination and tests specified in 4.7, 4.8.1, 4.8.2, 4.8.3, 4.8.4, 4.8.6, and 4.8.7.

4.5.2 Sampling for measurements and tests performed on sections cut from hose. Two hoses, identified as first sample hose and second sample hose shall be selected at random from the lot (see 4.5.1). Short lengths cut from one end of these hoses shall be used for the tests specified in 4.8.5, 4.8.8, 4.8.9, and 4.8.11. The samples for the tests shall consist of the following:

(a) One section not less than 32 inches in length, cut from the first sample hose, for the burst pressure test.
(b) One section not less than 12 inches in length, cut from the first sample hose, for the tensile pull test.
(c) One section not less than 52 inches in length, cut from the second sample hose, for the measurements of tube and cover thicknesses and the tests for the properties of the rubber components.

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4.5.3 Sampling for tests of friction rubber properties. Six vulcanized samples of the rubber friction used to impregnate the reinforcement plies shall be used for the friction rubber tests of 4.8.11.1, 4.8.11.2, and 4.8.11.4. The samples shall be 6 by 6 inches by 0.080 ± 0.010 inch, and shall be accompanied by a written certification from the contractor that the sample has the identical composition as the friction used in the hose and an equivalent cure (see 6.2.2).

4.6 Rejection based on quality conformance inspection.

4.6.1 Rejection on basis of visual examination and measurements of length. The entire lot shall be rejected when the number of defective hoses exceeds the acceptance number specified in MIL-STD-105. In any case, every hose containing one or more major defects shall be rejected and counted against the acceptance number for defective hoses.

4.6.1.1 Classification of visual defects and defects in length. These defects shall be classified in accordance with the following:

(a) Tubes not meeting the requirements of 3.9.1 shall be considered to have one or more major defects. Other major visual defects shall be classified in accordance with FED-STD-162.
(b) Length defects, if consisting of deviations between one and three percent of the ordered length (see 6.2.1), shall be considered as minor defects in accordance with 4.6.1. The entire lot shall be rejected if the number of hoses having length defects between one and three percent of the ordered length exceeds specified AQL (see 4.5.1). The two hoses which are residual lengths from samples for tests (see 4.5.2) shall not be considered defective.

4.6.2 Rejection on basis of wall thickness. The entire lot shall be rejected if the wall thickness of any sample hose (see 4.5.1) is less than the minimum tolerance specified in table I. The entire lot shall also be rejected if the wall thickness of any sample hose is more than 0.016 inch above the maximum tolerance specified in table I. The lot shall be accepted if the wall thickness of any sample hose is above the maximum tolerance, provided that:

(a) The excess above the maximum tolerance is not more than 0.016 inch.
(b) The excess thickness prevails over not more than 25 percent of the circumference.
(c) An equal or greater proportion of the circumference has a wall thickness at least 0.016 inch under the maximum tolerance.

4.6.3 Rejection on basis of other tests performed on entire hose. The entire lot shall be rejected if any sample hose fails to meet the requirements for unit weight, proof pressure, or dimensional stability.

4.6.4 Rejection on basis of tests performed on hose sections and rubber sample. The entire lot shall be rejected if the hose sections (see 4.5.2) fail to meet the requirements for burst pressure, tensile pull, tube, or cover thickness, construction, or properties of the rubber components.

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4.7 Visual examination. The hose section of 4.7.1 shall be examined for conformance to the required construction of 3.1 and 3.9.1. Interior and exterior surfaces and the ends of sample hoses shall be examined for the visual defects described in FED-STD-162 and 3.9.1. A borescope shall be used to inspect the tubes of the sample hoses.

4.7.1 First article specimen. A hose section, 48 inches in length, shall be cut from the center of the first article hose and cut open longitudinally to expose the tube surface and shall be inspected in accordance with 4.7.

4.8 Tests. Unless otherwise specified herein, tests shall be performed at 80 ± 9 degrees Fahrenheit (°F).

4.8.1 Length. Lengths of sample hoses shall be measured as specified in ASTM D 380.

4.8.2 Inside diameter. Inside diameters of sample hoses shall be measured as specified in ASTM D 380 using plug gauges.

4.8.3 Wall thickness. Wall thicknesses of sample hoses shall be measured at both ends of the hose using a vernier caliper. Measurements shall be made at least every 90 degrees around the circumference.

4.8.4 Unit weight. The unit weights of sample hoses (see 4.5.1) shall be determined by weighing the uncut hoses. The readings shall be accurate to the closest pound.

4.8.5 Tube and cover thicknesses. The thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.5.2(c). The thicknesses shall be measured in accordance with ASTM D 380. The respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which shall meet the requirement. The respective values for the cover thicknesses shall be treated in the same manner. A minimum of three points on each layer shall be tested.

4.8.6 Proof pressure. Each of the sample hoses shall be equipped with couplings in accordance with MIL-C-24356 and subjected to the proof pressure specified in 3.5. Before starting the test, a 10-pound hydrostatic pressure shall be established in the hose, and the length, outside diameter, and zero index for the twist shall be measured in accordance with ASTM D 380 in preparation for the dimensional stability test specified in 4.8.7, except that the hydrostatic pressure rate shall not exceed 1000 lb/in^2 per minute. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5-minute period and while the pressure is maintained, the length, outside diameter, and twist of the hose shall be measured for the determination of dimensional stability.

4.8.7 Dimensional stability. The effect of the proof test pressure (see 3.5) on the length, outside diameter, and twist of all sample hoses shall be determined as specified in 4.8.6 and ASTM D 380. The pressure shall then be reduced to zero lb/in^2 and the hose left in this condition for 5 minutes. At the end of this period, the diameter and length of the hose shall be measured.

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4.8.8 Burst pressure. The hose section obtained in accordance with 4.5.2(a) shall be equipped with couplings in accordance with MIL-C-24356. The burst pressure of this section in the straight condition shall then be tested in accordance with ASTM D 380.

4.8.9 Tensile pull. The hose section obtained in accordance with 4.5.2(b) shall be equipped with couplings in accordance with MIL-C-24356. The assembly shall then be subjected to the tensile load as specified in 3.8, which shall be maintained for not less than 5 minutes. Upon release of the tensile load, the hose section shall be tested to the proof pressure of table I in accordance with 4.8.6.

4.8.9.1 Elongation and neckdown (type B). One complete 36 inch to 48 inch length of type B hose shall be subjected to a tensile pull of 6,000 pounds. The elongation and neckdown shall conform to 3.8.1.

4.8.10 Kink stress. Prepare a 48-inch long hose section from the first article length with couplings in accordance with MIL-C-24356. Blank off the ends of the couplings, fill the hose section completely with type III fluid of TT-S-735 and pressurize to 50 lb/in^2. Hold the 50 lb/in^2 pressure for 48 $\pm$ 1/2 hours. At the end of the 48 hours, depressurize and drain the hose. Bend the hose over a 2-inch pipe such that the crease in the hose is located 1 foot from a coupling and secure that coupling to a vertical wall or frame so it will not move; see figure 1. At the other end of the hose section, attach and suspend a 650 lb weight for not less than 15 minutes. At the end of the 15 minutes, remove the weight, fill the hose completely with water and apply a proof pressure of 300 lb/in^2 to the hose. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5 minutes, depressurize and drain the hose. Remove the couplings and cut the hose section lengthways. Examine the hose for evidence of tube, interlayer adhesion or reinforcement failure. Evidence of leakage, noticeable weakness, or failure of adhesion or reinforcement shall be cause for rejection of the first article.

4.8.11 Properties of rubber components. The properties of rubber components, except testing of the friction rubber, shall be determined on longitudinal specimens prepared from the hose section obtained in accordance with 4.5.2(c). After separating and buffing smooth, the tube and cover samples shall rest not less than 30 minutes before preparing test specimens. The testing of the friction rubber shall be determined using specimens cut from the sample obtained in accordance with 4.5.3.

4.8.11.1 Initial tensile properties. Initial tensile strength and ultimate elongation shall be in accordance with ASTM D 412 using specimens cut with die C. The thickness of the die C specimen shall be 0.080 $\pm$ 0.010 inches.

4.8.11.2 Tensile properties after immersion. The tensile strength and ultimate elongation after immersion in type III fluid of TT-S-735 shall be in accordance with method 6111 of FED-STD-601, using specimens cut with die C of ASTM D 412. Tensile strength shall be based on the swollen cross-sectional area of the specimen. The thickness of the die C specimen shall be 0.080 $\pm$ 0.010 inches.

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4.8.11.3 Tensile properties after oven aging. The tensile strength and ultimate elongation after oven aging shall be determined in accordance with ASTM D 573 using specimens cut with die C of ASTM D 412, except that the aging period shall be 70 ± 1/4 hours at 212 ± 2°F. The thickness of the die C specimen shall be 0.080 ± 0.010 inches.

4.8.11.4 Volume change after immersion. The change in volume in percent of the original volume after immersion in type III fluid of TT-S-735, shall be determined on samples of the tube, cover, and friction in accordance with method 6211 of FED-STD-601.

4.8.11.5 Initial adhesion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with ASTM D 413, Machine Method and shall meet the requirements of 3.9.

4.8.11.5.1 Specimens. Strip or ring specimens shall be prepared in accordance with ASTM D 380, except that the specimen shall be a minimum of 8 inches in length for type A hose. For determining adhesion between plies of reinforcement, the plies to be separated must be at opposing angles. Three specimens of each interface shall be tested.

4.8.11.6 Adhesion after immersion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined after the test specimens (see 4.8.11.5.1) have been immersed in type III fluid of TT-S-735 for 46 ± 1/2 hour at 80 ± 9°F. The test procedure for adhesion shall be in accordance with ASTM D 413, Machine Method, except that the wet specimen shall be tested within 15 minutes after it is removed from the fluid. The adhesion values shall meet the requirements of 3.9.

4.8.11.7 Nonvolatile extractable material.

4.8.11.7.1 Test specimen. Test specimen shall consist of strips about 1/16-inch square by 2 inches long, totaling about 5 grams (g) in weight, cut from buffed pieces of the tube of the hose.

4.8.11.7.2 Procedure. Specimen shall be weighed and placed in an extraction thimble in the extracting apparatus in accordance with ASTM D 297. One hundred milliliters (mL) of type I fluid of TT-S-735 shall be added to the flask. The specimen shall be extracted for 46 ± 1/2 hour at such a rate that 2-1/2 to 3-1/2 minutes are required to fill and empty the extraction cup. The extract shall be poured into a weighed container. The specimen and flask shall be rinsed with 25 mL of fresh type I fluid and the wash added to the extract. The nonvolatile residue shall be determined from the extract in accordance with ASTM D 381, except that the evaporation time shall be 45 minutes. The weight of the residue shall be expressed as a percent of the original weight of the specimen. The test shall be performed in duplicate and the results averaged.

4.8.11.8 Phosphate plasticizer test. Phosphate plasticizer shall be determined qualitatively in accordance with method 15825 of FED-STD-601, using a small piece of clean sodium about the size of a pea, a test tube about 3 inches long, and the specimens of the residue remaining after the nonvolatile extractable material test specified in 4.8.11.7. The procedure shall be

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repeated if phosphorus is found, omitting the addition of the nonvolatile extractable material in order to determine whether phosphorus was present in any form in any of the reagents used.

4.8.11.9 Resistance to ozone.

4.8.11.9.1 Test specimens. Four 5 by 1 inch specimens of cover material shall be cut from the sample section of hose. The thickness of the specimens shall be 0.080 ± 0.010 inches. Two specimens shall be cut along the longitudinal direction of the hose and the other two shall be cut in the transverse direction of the hose.

4.8.11.9.2 Apparatus. Apparatus shall consist of a test chamber with an ozone generating source outside the chamber maintaining an ozone-air atmosphere in the chamber at the temperature and concentration desired; a circulating air oven, as described in ASTM D 573 and two pairs of metal plates, 3/8 by 1-1/2 by 8 inches each, with bolt holes spaced centrally 1/2 inch from each end, and provided with bolts and nuts for clamping the specimens.

4.8.11.9.3 Procedure. Specimen shall be placed between the metal plates and clamped so that the distance across one of the loops thus formed is 2 inches. The bulk of the specimen shall be reduced by cutting away the loop that projects from the side opposite the measured loop. The cut edges of the specimen shall be coated with paraffin to seal them from ozone. Figure 2 is a drawing of the clamped specimen. The clamped specimens shall be conditioned in an air oven for 16 to 24 hours at 100 ± 2°F. The specimens shall then be placed in the ozone test chamber for 166 ± 1/2 hours at 105 ± 2°F. The concentration of ozone in the chamber shall be 1.00 ± 0.05 parts of ozone per one million parts of air by volume. Specimen shall be examined at the end of this period for cracking of the hose cover using 7X magnification to determine conformance with the requirements specified in 3.9.

4.8.11.10 Resistance to water. Type C hose only.

4.8.11.10.1 The inner tube surface of one hose sample per lot shall be extracted in distilled water at reflux temperature in accordance with the following procedure:

(a) Remove a 1-inch ring of rubber inner tube and cut or buff away the outer surface until the surface is smooth.
(b) Cut sufficient length of inner tube to obtain approximately 10 g, based on a minimum thickness of 3/64 inch.
(c) Subject sample to distilled water, at reflux temperature, for 7 hours in an extraction apparatus as shown in method 16001 of FED-STD-601.
(d) Filter the solution through No. 40 Whatman filter paper or equivalent and collect the extract in a tared container.
(e) Extract the rubber sample in an additional 50-75 cubic centimeters of distilled water at a reflux temperature for 2 more hours and repeat step (d). Combine the extract solutions and evaporate to dryness. The container and residue shall be dried in an oven at 221 ± 5°F for 1 hour, cooled in a desiccator and

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weighed. The weight of the residue shall be record. A blank shall be run using the same amount of distilled water. After making allowance for the blank, the weight of the residue shall be recorded to the nearest milligram.

(f) Report the amount of matter extracted from the rubber in milligrams per square inch of surface, determined as follows:

Length (inches) x width (inches) x 2 + total surface area (square inches).

Milligrams per square inch = weight of extract/total surface area.

4.8.11.11 Specific gravity. The density of tube, friction, and cover material shall be measured in accordance with the Hydrostatic Method of ASTM D 297. The value shall be converted to the equivalent specific gravity and recorded.

4.8.11.12 Hardness. The durometer hardness of tube, friction, and cover material shall be measured and the values recorded in accordance with ASTM D 2240.

4.9 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing, and marking. Hose shall be furnished in the form of coil, reel, or straight lengths as specified (see 6.2.1). Hose in the form specified shall be preserved-packaged level A or C; packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-P-775.

5.2 Hose ends. Open ends of hose, with or without fittings shall be sealed against the free circulation of air using barrier material as specified in MIL-P-775.

6. NOTES

6.1 Intended use. Hose is intended for use in, but not limited to, refueling-at-sea. The maximum operating pressure of hoses is 150 lb/in^2.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Types and sizes (see 1.2).
(c) Length and size of hoses required (see 3.3) and whether or not residual lengths will be accepted.

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(d) When length defects consist of deviations between one and three percent of the ordered length of hose (see 4.6.1.1).
(e) Level of preservation-packaging and level of packing required (see 5.1).
(f) Form required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.9.1.1 and 4.5.3	Certificate of compliance	DI-E-2121	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Suggested formulations. Suggested formulations for the tube, friction and cover stock of the hose are listed as follows:

Tube stock

Compounding ingredients	Parts by weight
Paracril CLT	100
Philblack N330	60
Sulfur	1.5
Protox 166	5
Stearic acid	1
Hycar 1312	10
TP-90B	10
Altax	1
Methyl tuads	0.2

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Friction stock

Compounding ingredients	Parts by weight
Hycar 1041	100
Philblack N550	10
Hi Sil 233	30
Dixie clay	30
Sulfur	2
Protox 166	5
Stearic acid	1
Cumar P-10	30
Bardol B	15
Hylene MP	10
Altax	1.5

Cover stock

Compounding ingredients	Parts by weight
Paracril OZO	100
Philblack N330	30
Regal 300	30
Sulfur	1.5
Protox 166	5
Stearic acid	1
Flexzone 6H	3
Cumar P-25	15
Plasticizer SC	10
Flexol DOP	10
Altax	1.5
Thionex	0.2

6.3.1 Certain ingredients appear as proprietary names in the formulations listed in 6.3 since they were the specific ingredients used in the development of this specification. It is not intended to limit the choice of the commercial source of an ingredient or to infer that one source is better than another.

6.3.2 A hose constructed using the rubber formula of 6.3 has been found to meet the requirements of this specification. It is emphasized that this information is offered only as a suggestion. The Government does not guarantee that hoses made using these formulations will conform to the requirements of the specification. Use of part or all of this information for manufacturing hoses will not relieve the contractor from having his product tested for conformance with the specification.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 Subject term (key word) listing.

Adhesion
Non-volatile extractable material
Ozone, resistance to
Phosphate plasticized
Tube stock

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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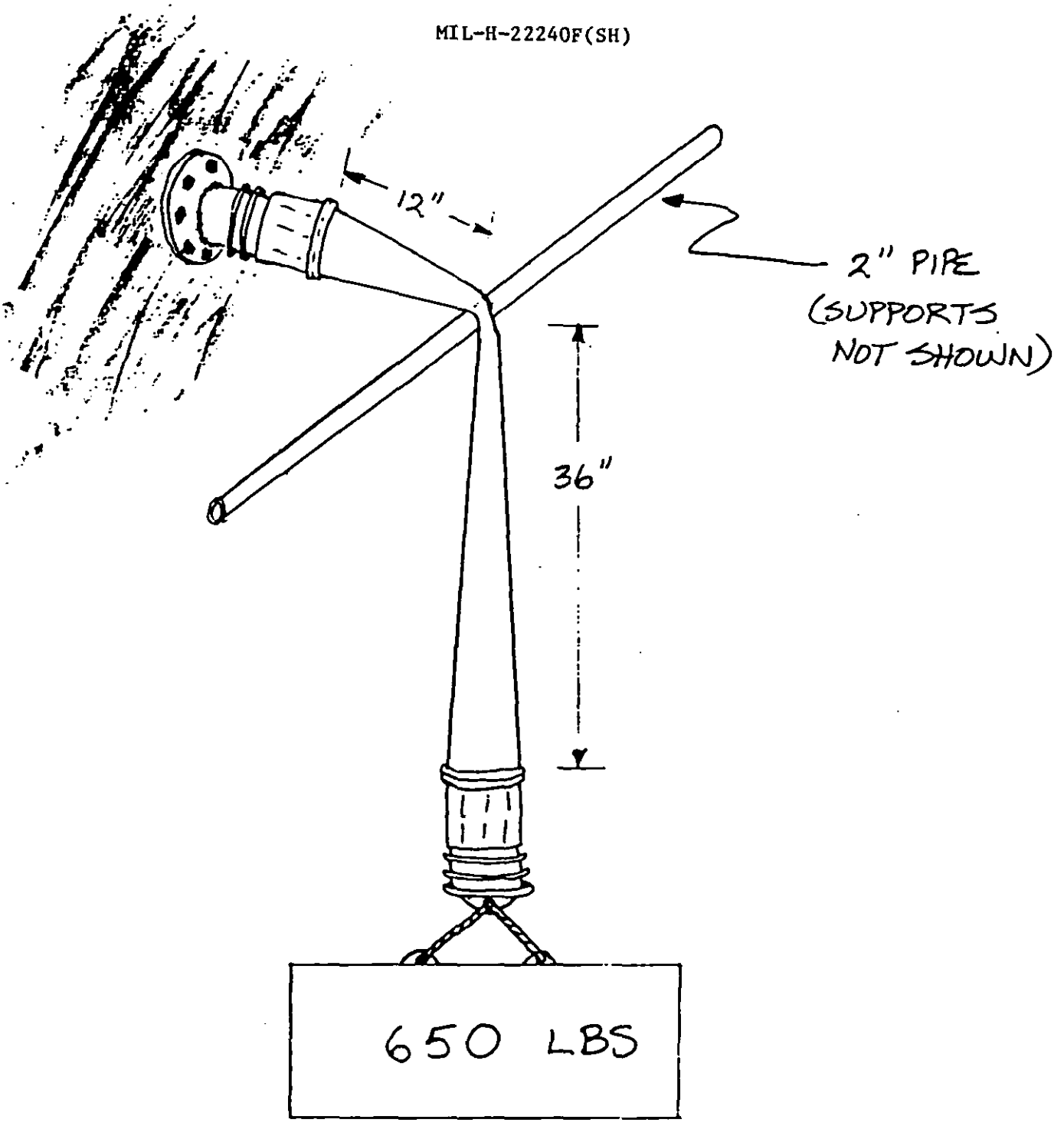


FIGURE 1.

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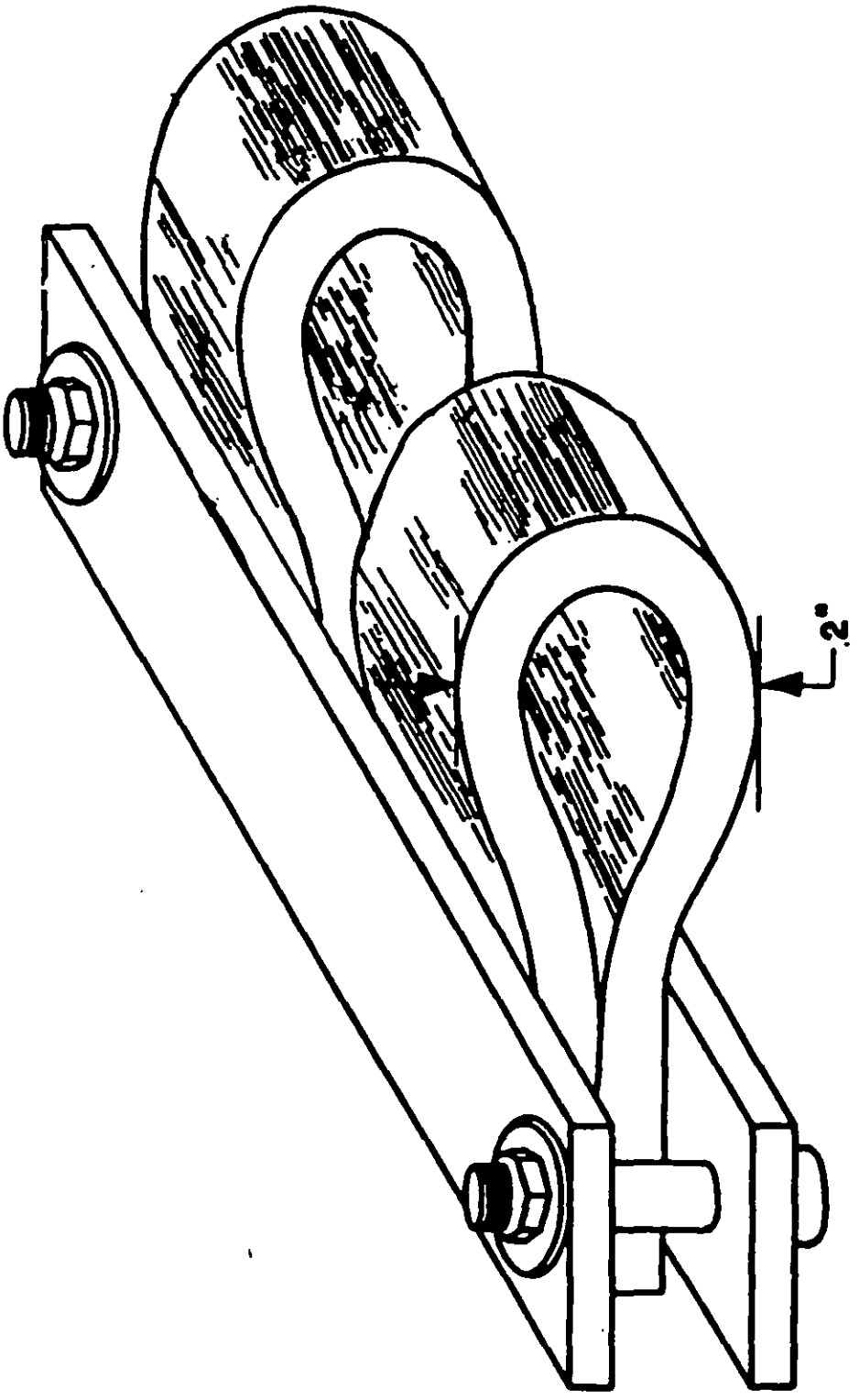


FIGURE 2.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-22240F(SH)	HOSE, RUBBER, PETROLEUM BASED FUELS AND WATER SERVICES, DISCHARGE ONLY, SMOOTH BORE, LIGHTWEIGHT BUOYANT TYPE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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